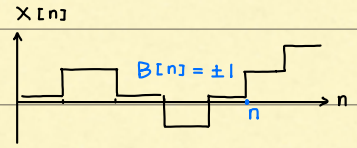
RW (Random Walk)



$B[n] = 1$ or -1 with $\frac{1}{2}$

DTRW $X[n] = \sum B[i]$ $\quad D(X[n]) = n$, $\quad R[n_1, n_2] = \min(n_1, n_2)$

martingale $\quad$ Pf: $R[2,3] = E[(B_1 + B_2)(B_1 + B_2 + B_3)] = 2$

cross term $= 0$, $E(B_1^2) = \frac{1}{2}\cdot 1^2 + \frac{1}{2}\cdot(-1)^2 = 1$

e.g. $X_n \sim Ber(\frac{1}{3})$, $Y_n = \sum_{i=1}^{n} X_i$, Find $C_Y(t_1, t_2)$

$E(X) = \frac{1}{3}$, $\quad E(X^2) = \frac{1}{3}$, $\quad D(X) = \frac{2}{9}$

$E(Y_n) = \frac{1}{3}n$

$R_Y(n_1, n_2) = E(Y_{n_1} Y_{n_2})$ $\qquad (X_1 + X_2)(X_1 + X_2 + X_3)$

共 $n_1 n_2$ 项，$\min(n_1, n_2)$ 项 $E(X^2)$，剩下的是 $E(X_1 X_2) = E(X_1)E(X_2)$

$= \min(n_1, n_2)\cdot\frac{1}{3} + [n_1 n_2 - \min(n_1, n_2)]\cdot\frac{1}{9}$

$C_Y(n_1, n_2) = R_Y(n_1, n_2) - \frac{1}{3}n_1 \cdot \frac{1}{3}n_2$

$= \frac{2}{9}\min(n_1, n_2)$

Wiener Process

$X(0) = 0$, $\quad X(t)$ continous

Increment indp $\quad X(t_1) - X(t_2)$ indp with $X(t)$, $t \le t_1$

Increment stationary $\quad X(t_1) - X(t_2)$ only about $t_2 - t_1$

$X(t) \sim N(0, t)$ $\qquad$ general $X(t) \sim N(0, \sigma^2 t)$

Prop. $R(t_1, t_2) = \sigma^2 \min(t_1, t_2)$

$X(t_1) - X(t_2) \sim N(0, t_2 - t_1)$ indp with $X(t_1)$

Wiener Process is Gaussian Process

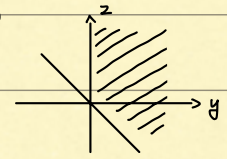
$$\begin{bmatrix} X(t_1) \\ \vdots \\ X(t_n) \end{bmatrix} \sim N\left(0, \begin{bmatrix} t_1 & t_1 & t_1 \\ t_1 & \ddots & \vdots \\ t_1 & \cdots & t_n \end{bmatrix}\right)$$

e.g. $X(t)$ Wiener Process, find $P(X(1) \geq 0 \cap X(2) \geq 0)$

$$P(\underbrace{X(1)}_{Y} \geq 0 \cap \underbrace{X(2) - X(1)}_{Z \sim N(0,1) \text{ indp with } X(1)} \geq -X(1))$$

$Y, Z \sim N(0,1)$, indp

$$P(Y \geq 0 \cap Z \geq -Y) = \frac{3}{8}$$



e.g. Find $P(X(1) + X(2) + X(3) > 2)$

$$\begin{bmatrix} X(1) \\ X(2) \\ X(3) \end{bmatrix} \sim N\left(0, \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}\right) \qquad Y = (1 \ 1 \ 1) \begin{bmatrix} X(1) \\ X(2) \\ X(3) \end{bmatrix} \sim N(0, 14)$$

$$P(Y < 2) = \Phi\left(\frac{2}{\sqrt{14}}\right)$$

WGN (White Gaussian Noise)

WGN is derivative of Wiener

$$\text{Wiener} \xrightarrow{X} \boxed{\frac{d}{dt}} \xrightarrow{Y} \text{WGN}$$

$X(t) \sim N(0, \sigma^2 t)$ $\qquad$ $Y(t) \sim N(0, \sigma^2)$

$R_X(t_1, t_2) = \sigma^2 \min(t_1, t_2)$ $\qquad$ $R_Y(t_1, t_2) = \sigma^2 \delta(t_1 - t_2)$ $\qquad$ Pf : $R_Y = L_{t_1} L_{t_2} R_X$

$S_Y(\omega) = \sigma^2$

WGN 的 spectral 表示为 $S(\omega) = \frac{1}{2} N_0$